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LAB MANUAL: INTERCONNECTING TWO SWITCHES WITH 6 PCS (SIMULATION)

TITLE

Connecting Two Switches with Multiple PCs and Testing using Simulation Mode in Cisco Packet Tracer

Objective

- To interconnect two switches in a LAN
- To connect six PCs across two switches
- To verify communication using **ICMP (Ping)** in Simulation Mode

Apparatus / Requirements

- 2 × 2960 Switches
- 6 × PCs
- Copper Straight-Through cables
- Cisco Packet Tracer

Theory

A switch is a networking device used to connect multiple devices in a Local Area Network (LAN). When two switches are connected, they allow communication between devices connected to different switches.

Switches forward data using **MAC addresses**. In this lab, **ICMP protocol (ping)** is used to test connectivity. Also, **Spanning Tree Protocol (STP)** runs in the background to prevent loops.

NETWORK TOPOLOGY

PROCEDURE

Step 1: Create Topology

- Open Cisco Packet Tracer
- Place:
    - 2 switches
    - 6 PCs

Step 2: Connect Devices

PC Connections:

- PC0 → Switch0 (Fa0/1)
- PC1 → Switch0 (Fa0/2)
- PC2 → Switch0 (Fa0/3)
- PC3 → Switch1 (Fa0/1)
- PC4 → Switch1 (Fa0/2)
- PC5 → Switch1 (Fa0/3)

Switch-to-Switch Connection:

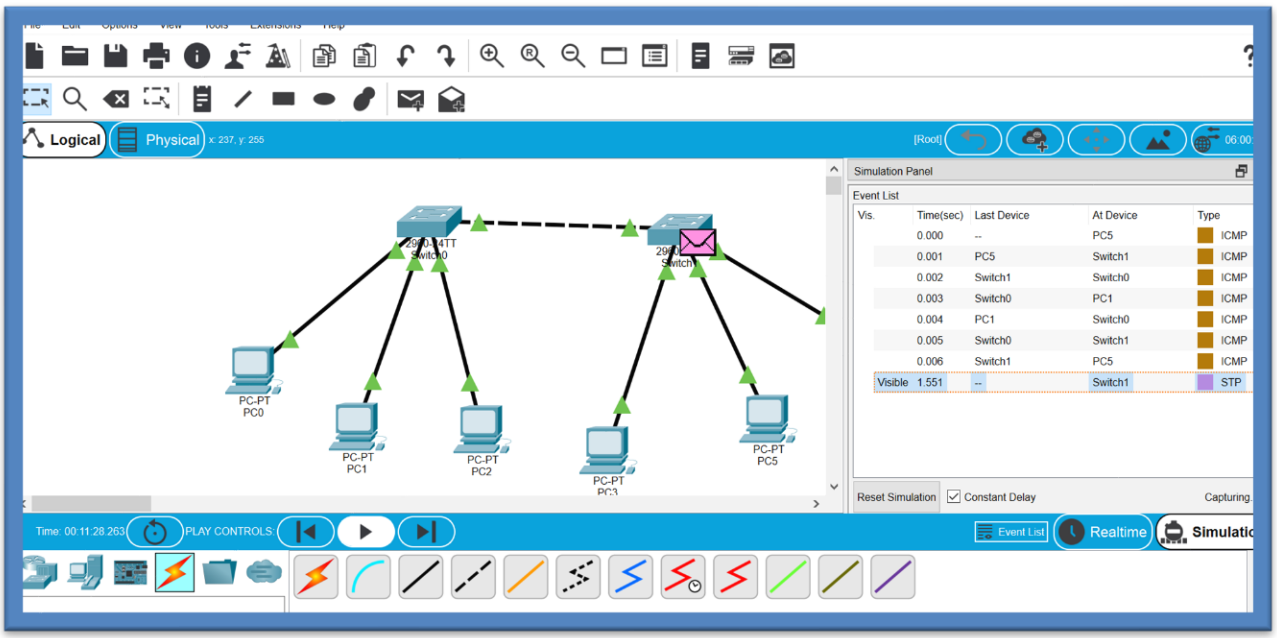
- Switch0 (Fa0/24) → Switch1 (Fa0/24)

Step 3: Assign IP Addresses

PC	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0

Step 4: Simulation Testing

1. Switch to **Simulation Mode**
2. Use **Add Simple PDU (envelope icon)**
3. Send packet from:
    - **PC5 → PC1** *(as shown in screenshot)*



Step 5: Observe Packet Flow

From Simulation Event List:

PC5 → Switch1 → Switch0 → PC1 → Switch0 → Switch1 → PC5

✓ This shows:

- Request reached destination
- Reply returned successfully

Observations

- Successful ICMP packets confirm connectivity
- Data travels through both switches
- **STP protocol** is visible in simulation (normal behavior)
- Green indicators show active links

Precautions

- Ensure all PCs are in same network
- Use correct cable type
- Wait for links to turn green

- Avoid duplicate IP addresses

Result

The two switches were successfully interconnected, and communication between six PCs was verified using simulation mode and ICMP protocol.

CONCLUSION

The experiment demonstrated that multiple PCs connected to different switches can communicate successfully when switches are interconnected, and data transmission was verified using ICMP in simulation mode.